

Optimized and Reconfigurable Hardware Design for ASCON AEAD and Hash Standards

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Abstract—NIST has finalized LWC standardization process by selecting ASCON as the new standard. ASCON is a versatile algorithm supporting two primary functions: AEAD and HASH. The standard includes three AEAD variants “*ASCON-128*, *ASCON-128a*, *ASCON-80pq*” and two HASH variants “*Hash*, *Hasha*”. In this work an efficient programable hardware solution that supports all the three AEAD and two HASH variants is designed targeting GF12nm ASIC technology and Spartan-7 FPGA. The programmable design is compared to individual hardware implementations of each variant, demonstrating significant enhancements. By utilizing a single programmable design instead of five separate ones, 61% reduction in total silicon area and 75% energy savings are achieved. In addition, a comprehensive comparison of hardware implementations for each ASCON variant is provided. Performance metrics such as area, throughput, and energy consumption are evaluated, revealing that ASCON-128a offers the lowest energy consumption (0.45 pJ/bit) with the highest throughput (12.15 Gbps), while ASCON-128 achieves the smallest silicon footprint (1242 μm^2 on GF12nm).

Keywords—LWC AEAD, ASCON, Energy, Cryptography, Programmable implementation.

I. INTRODUCTION

Current cryptographic standards have been standardized over two decades ago, these algorithms were selected based on their security strength with no limitations or constraints on their implementation resources. Therefore, these standards are not suitable in the IoT and Edge devices that have been rising in the recent years. To overcome such an issue, the National Institute of Standards and Technology (NIST) has evaluated and standardized a new Lightweight Cryptography algorithm (LWC) [1]. The standardization process concluded in 2023, selecting ASCON as the new LWC standard [2][3].

ASCON is a Lightweight Cryptography algorithm that supports two primary cryptographic operations, first one is Authenticated Encryption with Associated Data (AEAD). This operation combines authentication with encryption and decryption. The input, which can vary in length, is processed block by block, and the output is generated in the same manner, followed by an authentication tag. The main components involved in AEAD are *Key (K)*: The secret data in the process. *Nonce (N)*: Acts as an Initialization Vector. *Associated Data (AD)*: Additional data used for authentication. *Text (T) and Output (O)* which is either *Plain Text (PT)*: Data before encryption or after decryption. *Cipher Text (CT)*: Data after encryption or before decryption. Finally, *Tag*: The message authentication code. The second cryptography operation supported by ASCON is HASH. This operation generates a fixed-length hash for a given message. The *Message input (M)* is divided into blocks for processing to generate the output *hash (H)*.

ASCON features three main AEAD variants: *ASCON-128*, *ASCON-128a*, and *ASCON-80pq*, along with two HASH variants, *Hash* and *Hasha* (detailed in section III). To adhere to the latest standards, it is recommended to incorporate all

five variants into the design. This can be accomplished either by creating five distinct designs, each optimized for a specific variant, or by developing a flexible, programmable design that can dynamically adapt to the required variant based on the application’s specific needs.

In this study, we develop a programmable hardware implementation capable of executing any of the ASCON AEAD/HASH variants. The design is implemented on both Global Foundry 12nm (GF12nm) and Spartan-7 FPGA. Additionally, a comparative study against standalone hardware designs for each variant is conducted. The primary goal of this investigation is to highlight the benefits and efficiencies achieved through the programmable design, as opposed to the traditional approach, which requires separate designs for each variant. This study also serves as a benchmark for quantifying the implementation resources needed for each variant HW. The evaluation of the hardware focused on area, throughput, and energy consumption (energy consumed per bit). The paper is organized as follows: Section II describes ASCON algorithm specifications and stages. HW design is illustrated in section III, and results are shown in section IV. Finally, section V concludes the paper.

II. ALGORITHM SPECIFICATIONS

ASCON was first introduced as a submission candidate in the CAESAR competition for lightweight AEAD algorithms in 2015 [4]. After announcing ASCON as the winner of the competition early 2019, ASCON was submitted as a candidate for NIST LWC standardization. Early 2023, ASCON was selected as the new standard for LWC. ASCON AEAD and HASH top level and algorithm specifications are shown in figure 1 [3].

ASCON AEAD (figure 1.a) consists of four stages. **Initialization** where the state is initialized, **Process AD**: AD block is XORed with the state and permuted for “b” rounds. **Process T**: The state is updated by absorbing T (PT/CT) and generating O (CT/PT) based on the operation of encryption or decryption, and finally **Finalization** where Tag is generated for authentication. ASCON HASH (figure 1.b) has three stages. **Initialization** to initialize the state, **Absorb M** to absorb

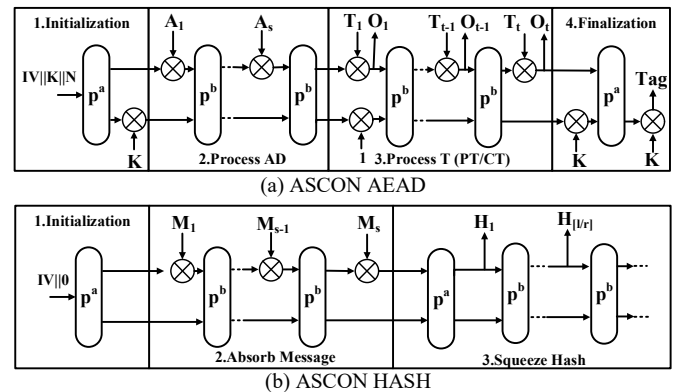


Fig 1. ASCON AEAD and HASH stages. AEAD has four main stages, Initialization, Process AD, Process T and Finalization, while HASH has three stages, Initialization, Process Message and Generate Hash.

TABLE I. ASCON VARIANTS, EACH VARIANT HAS ITS OWN SPECIFICATIONS.

Parameter	Variant	Block Size				Rounds Number	
		Key	Nonce	Tag	Data Block	<i>a</i>	<i>b</i>
AEAD	ASCON	128	128	128	64	12	6
	ASCONa	128	128	128	128	12	8
HASH		Hash			Data Block	<i>a</i>	<i>b</i>
	Hash	256			64	12	12
	Hasha	256			64	12	8

message blocks to generate the hash during *Squeeze Hash* stage [3].

III. ASCON VARIANTS HW DESIGN

In ASCON, each AEAD and HASH variant has its own characteristics. Table I describes the variants, showing data block size, number of rounds, and key length for each variant. Multiple work in literature implemented LWC accelerators in HW focusing on ASCON AEAD operation [5][6][7][8][12][13]. In addition, different programmable designs are also proposed such as [9][10][11]. In this section HW design of each variant in addition to the programmable design are detailed. All ASCON variants share the core *ASCON* round as a fundamental building block [3]. Each variant design includes a dedicated *controller state machine*, which is integrated on top of the accelerated *ASCON* round. The complexity of the state machine is adjusted for each variant by modifying the number of states to control looping rounds. ASCON variants details are described below:

A. AEAD variants

Main variant of ASCON AEAD is ASCON-128 (also referred as *Ascon*). It is considered to have the simplest implementation and HW requirements along the variants. The controller has 36 states “12 *Initialization*, 6 *AD block processing*, 6 *T block processing* and 12 *Finalization*”. Second AEAD variant is ASCON-128a (also referred as *Ascon-a*). It differs by having larger input block of 128 bits compared to 64 bits in ASCON-128 variant. Also, more rounds “8” are needed to permute the state after absorbing message blocks resulting in a more complex controller. As ASCON-128a variant requires more state rounds for AD and T blocks processing, the controller has a total of 40 states “12 *Initialization*, 8 *AD block processing*, 8 *T block processing* and 12 *Finalization*”. Last AEAD variant is ASCON-80pq (also referred as *Ascon-pq*) which shares the same specifications as *ASCON-128* variant but has a larger key size

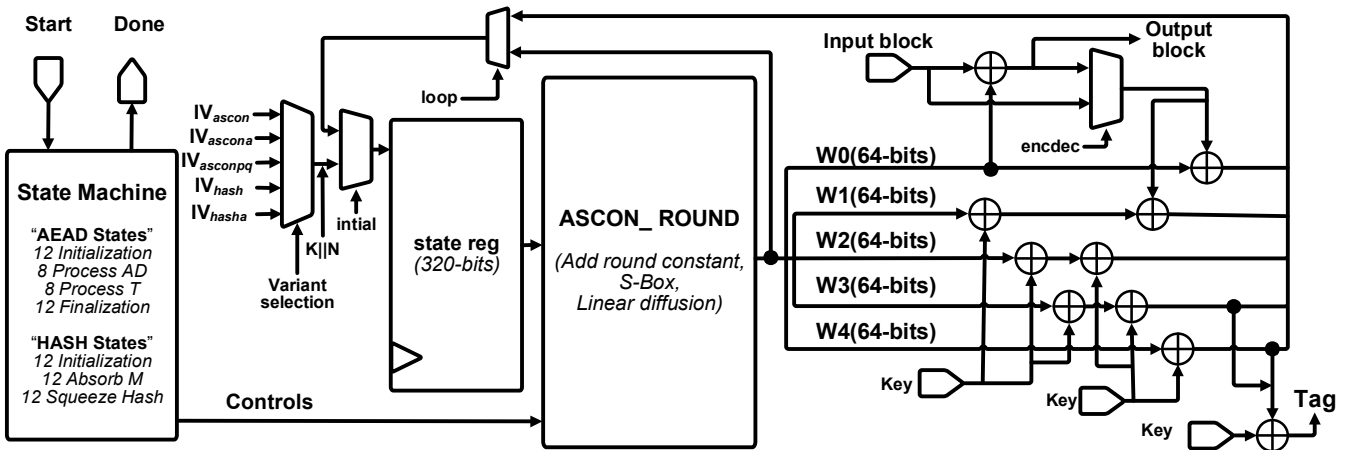


Fig 2. Programable design datapath, ASCON Round is instantiated along with state register to store the state. 5 different initialization vectors are provided based on the variant selected. The state machine has both AEAD and Hash states that controls the operation.

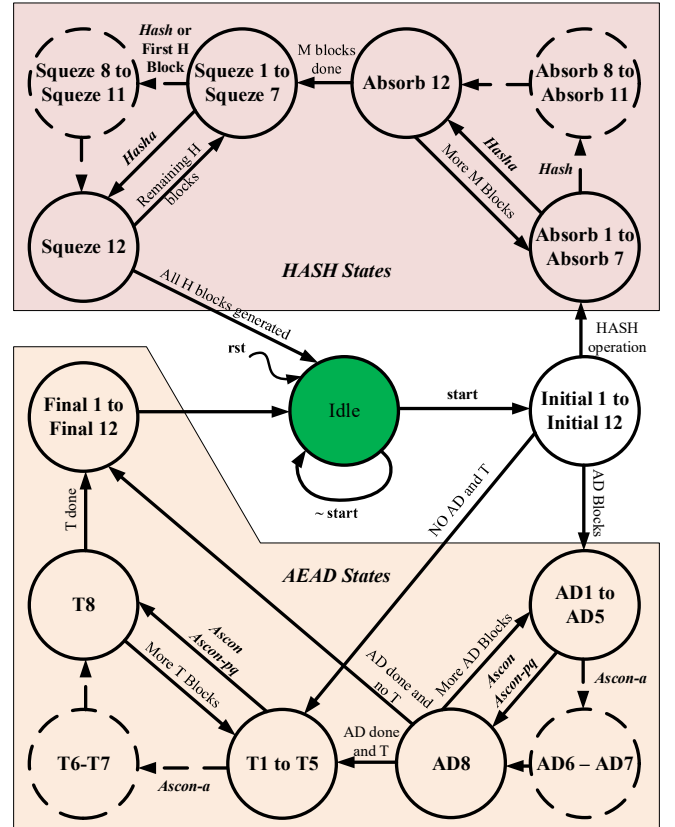


Fig 3. Programable ASCON state machine. It has two sections, AEAD states for AEAD variants, and HASH states for Hash variants.

of 160 bits instead of 128 bits to increase the search space of key and hence increase security in post-quantum era [3].

B. HASH variants

ASCON has two HASH variants that have the same datapath HW excluding key and tag. Moreover, the controller is different where *Hash* variant loops for 12 rounds during *M Absorb* stage, while *Hasha* variant needs 8 rounds. The controller of *Hash* variant consists of 36 states “12 *Initialization*, 12 *Absorb M* and 12 *Squeeze Hash*”. *Hasha* variant controller has a simpler controller with 32 states “12 *Initialization*, 8 *Absorb M* and 8-12 *Squeeze Hash*”.

C. ASCON_Programable

The final design presented in this study is a programmable architecture capable of handling all five ASCON variants. Since all variants share the same underlying hardware [3], the

design leverages resource sharing, allowing a single hardware platform to support all variants efficiently. The programmable architecture is shown in figure 2, it is designed including all gates required for each variant. Based on the variant needed, the input message, key size and rounds number are determined by the controller (that includes all states for all variants). Figure 3 shows the controller state machine for the programmable design, it has 64 states to cover all variants, states can be divided into two categories: AEAD (lower states) and HASH (upper states). Depending on the variant running, the controller runs through 36 AEAD states for *Ascon* and *Ascon-pq* variants (lower solid line states) or through 40 states for *Ascon-a* variant (lower solid and dash line states). For HASH operation, the initialization states are shared with AEAD initialization states, followed by Absorb states “12 (upper solid and dashed line states) for *Hash* or 8 (upper solid line states only) for *Hasha*”. H block is generated during Squeeze states that runs for 12 rounds in *Hash* variant or 8 rounds in *Hasha* variant (except for first H block that is generated after 12 rounds). Results shown in the next section presents the savings in HW by using the programmable design against having five separate designs for each variant.

IV. RESULTS

All designs including five ASCON standalone variants and programmable HW are designed and synthesized using Global Foundry 12nm (GF12nm) ASIC technology. In addition to ASIC, designs are synthesized over FPGA Spartan-7 to provide a comprehensive benchmark. The evaluation metrics are silicon area (ASIC) / resource utilization (FPGA), throughput, and energy consumption.

A. Area

GF12nm area of all variants and resource utilization over FPGA are shown in Table II. *Ascon* variant has the smallest area (1242 μm^2) among AEAD variants while *Ascon-a* is the largest requiring 1315 μm^2 . Both HASH variants have almost the same area which is 980 μm^2 . The programmable ASCON design has area of 1539 μm^2 , which equals approximately 1.56 times the area of *Hash*, and 1.24 times the area of *Ascon* in GF12 and Spartan-7. However, within a single programmable design it is possible to implement all AEAD and HASH variants with about 27% of silicon area as opposed to deploying five independent designs (5786 μm^2) as shown in Table II.

B. Throughput

The second metric in the evaluation is throughput. This is evaluated by using multiple scenarios for T and AD following strategy in [5]. AEAD has 6 scenarios include short (16B) and long (1536B) messages for "AD only," "T only," and "AD + T" cases. For HASH two input sizes are considered: short input (16B) and long input message (1536B). Short

TABLE II. ASCON VARIANTS AREA OVER GF12 AND SPARTAN-7.

Variant	Area (μm^2) (GF12nm)	Equivalent gates (KGE)	Utilization (LUT/FF) (Spartan-7)
<i>Ascon</i>	1242	8.56	805/362
<i>Ascon-a</i>	1315	9.06	966/366
<i>Ascon-pq</i>	1268	8.74	850/362
<i>Hash</i>	978	6.74	602/364
<i>Hasha</i>	983	6.77	603/368
<i>Sum of HW</i>	5786	39.86	3826/1822
<i>Programmable HW</i>	1539	10.60	1481 / 407
<i>Ratio</i>	27%	27%	39% / 22%

Resource constraint devices spectrum

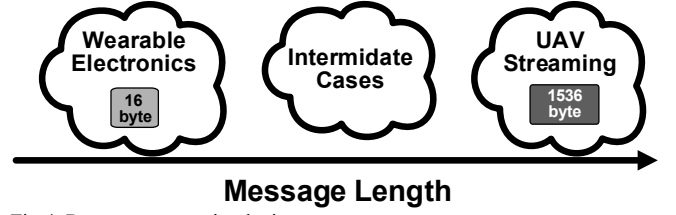


Fig 4. Resource constraint devices spectrum.

TABLE III. ASCON VARIANTS THROUGHPUT FOR DIFFERENT CASES ON GF12NM TECHNOLOGY.

HW	Throughput (Gbps)						Average
	1. Short AD	2. Short T	3. Short AD+T	4. Long AD	5. Long T	6. Long AD+T	
<i>Ascon</i>	3.72	4.324	5.82	12.98	13.046	13.16	8.842
<i>Ascon-a</i>	3.8	4.706	6.4	19.14	19.34	19.56	12.158
<i>Ascon-pq</i>	3.72	4.324	5.82	12.98	13.046	13.16	8.842
<i>Hash</i>		1.862			6.5		4.181
<i>Hasha</i>		2.42			9.69		6.055

messages mimic use cases such as wearable devices, while long messages mimic applications like UAVs, etc. This helps to evaluate the corner cases of resource constrained devices as shown in Figure 4. Synthesis and analysis are done with a frequency of 1 GHz for GF12 and 133 MHz for Spartan-7.

Table III shows the throughput results of each variant over GF12nm ASIC technology. Among AEAD variants, *Ascon-a* variants achieved the best throughput with an average of 12.158 Gbps. Even though *Ascon-a* has more rounds for block processing, it takes larger block in both process AD and process T stages, which results in higher throughput. *Ascon* and *Ascon-pq* variants achieve the same throughput of 8.842 Gbps. For HASH variants, *Hasha* has almost 1.5 times the throughput of *Hash* variant, they achieve 6.055 and 4.181 Gbps, respectively. It is noticed that for cases of long messages (Cases 4-6 in AEAD and case 2 in HASH) the throughput is higher, this is due to minimization of session initialization and finalization overhead (stages 1 and 4 in figure 1.a, stage 1 in figure 1.b) and most of the processing time is for data (AD/T) processing (stages 2 and 3 in figure 1.a, stage 2 in figure 1.b). Figure 5 shows the average throughput for all variants over GF12nm ASIC and Spartan-7 FPGA.

C. Energy Consumption

The last evaluation metric is the energy consumption (J/bit). Energy consumption is evaluated by dividing power by the throughput as described in equation 1.

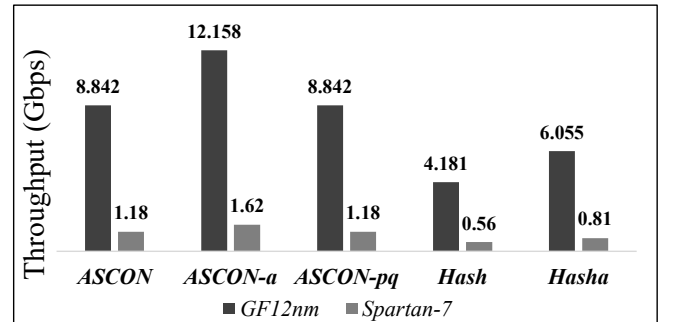


Fig 5. Average throughput of all variants over both platforms. GF12nm achieved ~7.5 times the throughput of Spartan-7 as it has a ~7.5x operating frequency. *ASCONa* variant has the highest throughput among AEAD variants and *Hasha* is the fastest HASH variant.

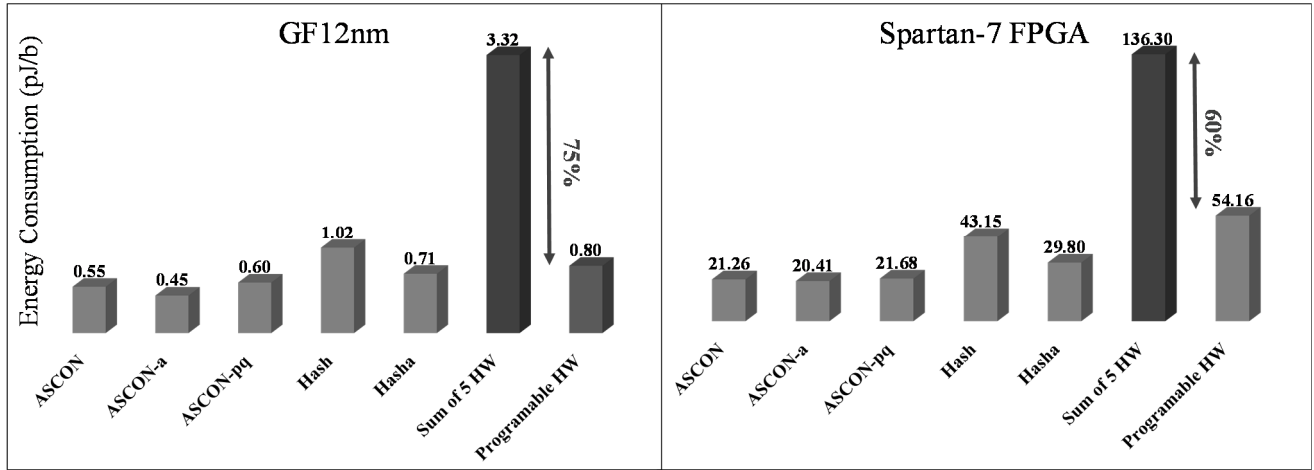


Fig 6. Energy consumption of running different variants over its standalone HW. Programmable HW energy consumption is shown and compared to total energy needed for 5 combined variant HW. Programmable HW consumes only 25% of energy compared to the 5 HW on GF12nm, and 40% on Spartan-7.

$$\text{Energy consumption (J/bit)} = \frac{\text{Power (J/sec)}}{\text{Throughput (bit/sec)}} \quad (1)$$

As a standard for LWC, hence, energy constraint is a main metric to evaluate. It determines the lifetime of a limited resource device battery. Figure 6 shows variants HW energy consumption compared to the programmable HW.

GF12 is more energy efficient compared to Spartan-7 as expected. Among AEAD variants *ASCONa* consumed the least energy with 0.45 pJ/b. For the main AEAD variant, *ASCON*, it consumes 0.55 pJ/b on GF12 and 21.26 pJ/b on Spartan-7. For HASH variants, *Hasha* has lower energy consumption compared to *Hash* variant. *Hasha* consumes 0.71 pJ/b compared to 1.02 pJ/b for *Hash* on GF12nm. Figure 6 illustrates the energy consumption of the programmable HW compared to the energy of the 5 variant HW combined. While the programmable HW is indeed less energy efficient on average compared to standalone designs, the programmable design saves 75% of the energy needed compared to the combined 5 variants HW on GF12nm and 60% on Spartan-7. On average, programmable HW energy consumption is 0.8 pJ/b on GF12nm and 54.16 pJ/b on Spartan-7.

D. State-of-The-Art comparison

Comparison of ASCON AEAD HW implementation with state-of-the-art is presented in Table IV. This work achieves

TABLE IV. ASCON VARIANTS COMPARISON WITH STATE OF THE ART.

	This Work	ISCAS'22 [5]	TCAS'24 [6]	ICECS'22 [7]	This Work	IoT-J'21 [8]
Technology	12nm	22nm	45nm	28nm	Spartan-7	Spartan-6
Variant	ASCON-128					
Area	1233 μm^2	2193 μm^2	N/A	4971 μm^2	805 LUT	2060 LUT
Throughput (Gbps)	8.842	0.351	1.57	N/A	1.18	0.315
Energy Consumption (pJ/b)	0.55	0.65	2.7	7.46	21.26	137
Variant	ASCON-128a					
Area	1315 μm^2	N/A	N/A	N/A	966 LUT	2930 LUT
Throughput (Gbps)	12.158	N/A	N/A	N/A	1.62	0.72
Energy Consumption (pJ/b)	0.45	N/A	N/A	N/A	20.41	103

TABLE V. PROGRAMABLE ASCON PERFORMANCE COMPARISON

	This Work		SOCC'22 [9]	ISDFS'24 [10]	PACET'24 [11]
Supported Variants	3 AEAD, 2 HASH variants		2 AEAD, 2 HASH variants	3 AEAD variants (No HASH)	2 AEAD, 2 HASH variants
Technology	12nm	12nm	28/32nm	22nm	22nm
Area	1539 μm^2	1539 μm^2	N/A	2670 μm^2	2558 μm^2
Case	Average cases 1 to 6	Long message only	Long message only	Average cases 1 to 6	N/A
Throughput (Gbps)	8	12.36	5.6	2	6.4
Energy Consumption (pJ/b)	0.8	0.42	0.41	0.56	N/A

lower area for *Ascon* and *Ascona* variants compared to other implementations in [5],[7] and [8]. Moreover, this work has higher throughput and more efficient energy consumption. This work achieves throughput that is 4 to 16 times the throughput reported, and 13.6 times lower energy consumption compared to [7] on ASIC and 5-6.4 times lower energy consumption compared to [8] on FPGA. For the programmable architecture, the comparison is shown in Table V. This work provides an architecture that supports 3 AEAD variants and 2 HASH variants compared to only 2 AEAD variants in [9] and [11] and no HASH in [10]. *Ascon-pq* variant is proposed for post-quantum era, it has a key size of 160 bits, hence it increases the search space for the secret key and gives a better post-quantum resistance, as described in [3]. This work achieves 4x throughput compared to [10] and 2.2x throughput compared to [9]. Energy consumption is almost the same as [9] for long messages, however the consumption is higher compared to [10], this is due to added variants of HASH that consume more energy and hence increase the average energy consumption. Nevertheless, this work has all 5 variants in less area with higher throughput.

V. CONCLUSION

NIST selected ASCON as the new LWC standard supporting AEAD and HASH functions. It is necessary to support both functions with all their variants. There are 3 AEAD and 2 HASH variants. This work includes the design and comparison between all ASCON variants hardware on both "GF12nm and Spartan-7 FPGA". Moreover, an optimized programmable design is proposed that supports all 5 variants, showing the advantages of the programmable design compared to integrating independent variants designs. Results show that *Ascon-a* variant is the most energy efficient AEAD variant while *Ascon* has the smallest area. Between HASH variants *Hasha* provides higher throughput and lower energy consumption. The programmable design saves ~63% area and reduces energy consumption by 75% compared to 5 independent variant implementations over CMOS GF12nm. In a comparison with previous research, this work provides a programmable design for all variants with lower energy consumption, higher throughput and smaller area.

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